

The Most Expensive Museum in the World: Three Player Cooperative Game Between VR and PC Platforms Investigating Empathy between Players and Historical Characters.

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Figure 1: Collage of the interactive experience, showing enlarged spider like robot in a virtual copy of control room of power plant Zwentendorf

ABSTRACT

The paper describes an art installation based on a local cooperative cross platform VR/PC game for three players. The story is based on the historical story of a nuclear power plant, which was built, but never turned on because of a public referendum. The

game mechanics are built upon the interaction of three playable historical characters: engineer, activist, and politician. The main goal was to foster empathy in the players by replaying the game from each character's perspective. We present the story implementation, custom-made interfaces, physical setup, and an evaluation of recordings from a four-month display in a public gallery.



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CCS CONCEPTS

• **Human-centered computing** → Collaborative interaction; Virtual reality; Keyboards; Haptic devices; • **Software and its engineering** → Interactive games; • **Applied computing** → Media arts.

KEYWORDS

Virtual Reality, Interactive Games, Media Art, Empathy, History, Serious Games, Crossplatform, Cooperation between PC and VR, Keyboard modification, Virtual reality physical modification

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1 INTRODUCTION

1.1 Historical background as a base for game characters

The narrative of the game is based on a real story about the first commercial nuclear power plant in Austria, located in a municipality Zwentendorf. The initiation of the power plant was prevented by a public referendum with a narrow majority of 50.47% against the startup which led to a total ban of use of nuclear energy in Austria with an exception for science [Pelinka 1983]. The nuclear energy debate across Europe stays unsettled, highlighted when Finland unveiled a new reactor shortly after Germany closed its plants [with AP 2023].

In the story of Zwentendorf, there were three major forces involved, represented by activists, engineers, and politicians. Some anti-activists like Peter Weisch or Friedrich Hundertwasser dreamed of setting up in the closed power plant a museum of obsolete technologies and call it “The Most Expensive Museum in the World.” [Koeb nd] More irrational anti-activists like Udo Proksch wanted to turn the power plant into a graveyard, where people would be buried in vertical position. This did not happen, as Udo Proksch was put in jail for insurance fraud with alleged uranium equipment. After he died, he was buried in a horizontal position [Zach 2019].

Some of the engineers offering a pragmatic explanation took part in debates as their arguments were hindered by Austrian bureaucracy and a general dissatisfaction of the public in relation to science [Hirsch and Nowotny 1977]. Another take on the engineering stance could be elucidated by the character of Johann Fleischer, a service engineer who was alone looking after the power plant after it was closed. Among his duties was a task to visit all the rooms in the power plant at least once a month [Fleischer 2018].

The political character could be understood as being the Austrian chancellor Bruno Kreisky, who first listened to engineers and decided to build the factory and then after it was built, he decided to run the referendum [Martinovsky 2012]. Although he said that he would resign if the referendum wouldn't be in favor of nuclear energy, he remained in political power long after that.

The referendum's aftermath persists. Post-closure, the power plant transformed into a training hub, temporary school, and venue for music festivals like Shutdown and Nuke, also appearing in the film “Meltdown” starring Dolph Lundgren [Parotte 2022]. Since 1983, Austria has sourced electricity from Czechoslovakia's Dukovany plant. The Dürnrohr coal plant, 2.5 km from Zwentendorf, was constructed in 1987 and later converted to a natural gas facility [press service 2022]. In April 2022, the Austrian Chancellor's visit

to Putin marked the first meeting between an EU leader and Putin since the Ukrainian conflict escalated [Shields 2022]. Lower Austria's flag closely resembles Ukraine's. Austria's gradual reduction of Russian gas continues, but with substantial ongoing payments to Russia [Jack 2023]. This story is the basis of our application, focusing on role playing of characters with opposing opinions.

1.2 Crossing the thresholds - how can interactive digital media foster empathy?

We aimed to foster empathy in players for characters with opposing opinions, inspired by the story of Zwentendorf. Based on historical events, we identified three key figures: an engineer, activist, and politician. Our goal was to foster empathy between players and their historical characters by the property of the game to end more easily in a bad end scenario, replay, switch roles and replay the same story from a perspective of the opposing character. The medium of a PC/VR game was chosen because the concept of re-playing, dying, and playing again as an opposing character is a common technique to induce empathy. We followed the example of Belman and Flanagan in “Designing Games to Foster Empathy”, which states: “*Games are well-suited to [fostering empathy] because they allow players to inhabit the roles of other people in a uniquely immersive way. One can read about Darfuri refugees in the news, but, in an admittedly limited sense, a game can allow one to be a Darfuri refugee.*” [Belman and Flanagan 2010]. Following that in our scenario, players keep memories of their experiences while playing as an opposing character. This allows them to interact with a deeper understanding of that character when they switch roles, ultimately enhancing their ability for empathy.

As argued by Hoy: “*the games show success in conveying empathy when compared to traditional approaches because they allow students to experience structural constraints and social norms first-hand*” Our approach was to induce empathy by allowing players to have first-hand experience with the historical characters. Often, the root of understanding other people's problems is a lack of empathy stemming from limited direct experience with the issues [Hoy 2018]. A historical story like this is full of contradictory stances, usually because it is much harder to sympathize with someone who has a completely different opinion, and we cannot understand the behavior until we become the character. With digital media, especially VR, becoming someone else is easier than before [Bachen et al. 2016]. In a serious game like this, it is recommended to prioritize inducing empathy and immersion [Tettegah et al. 2006].

The use of VR and empathy was investigated among cinematic experiences [Darnell and Hutchinson 2016], in relationship to physical and social presence [Felnhofer et al. 2014] or in prosocial VR games [Ho and Ng 2022]. The last example is important since the findings show that perspective-changing experiences indirectly enhance game immersion and induce empathy. Earlier studies suggested that identification with a character can enhance players' empathy and enjoyment [Hefner et al. 2007]. The effort behind this work focuses on the question of whether we can help people to see the world and their own actions from an opposing perspective by switching characters, and finally induce some empathy towards other opinions.

2 METHODS

2.1 Game mechanics, general goals of the experience and spatial organizations

The experience is designed as a role playing replayable multiplayer cooperative experience. Visitors can play as an engineer, activist or politician. They meet in a virtual copy of a control room in Zwentendorf. The players were physically divided by entering a physical niche in a gallery wall, which were specifically designed for the need of every character, as well as with a custom-made user interface. In the niche the visitors would find either a PC or a VR headset, allowing the users to enter a virtual copy of a Zwentendorf nuclear power plant control room as in Figure 2.

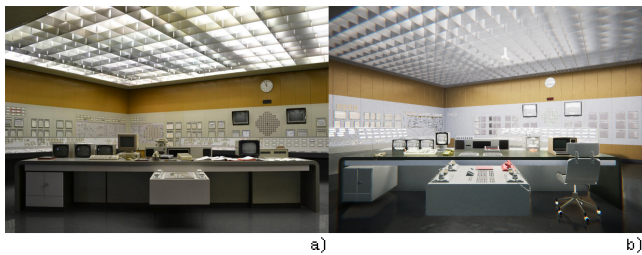


Figure 2: Comparison of a) The original control room (© Wikimedia Commons(https://commons.wikimedia.org/wiki/File:Kontrollraum_AKW_Zwentendorf_20121116.jpg) b) The virtual copy (© Vojtěch Radakulan)

The game mechanics are designed as shown in Figure 3. The engineer sees the activist as a red haze, and the activist sees the engineer as a robot. The politician can only see the room as if from an old CCTV camera. They all can talk to each other.

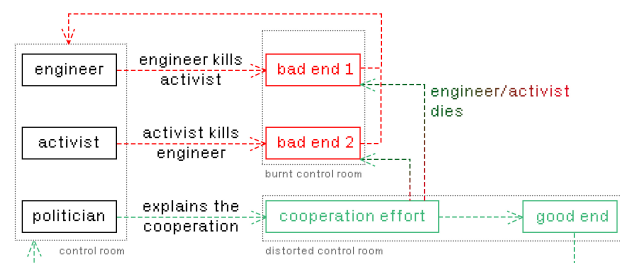


Figure 3: Game mechanics in a schematic graph (© Vojtěch Radakulan)

The engineer has an ability to extract power from reactors and is instructed to do so, which takes lives from activists. The activist has a gun, which is instructed to use against the engineer and the gun draws the engineers' lives.

There are two possible bad endings. If the engineer kills the activist, they appear in a burnt version of the control room, where the activist has full control of the engineer's robot, and they no longer communicate by voice. Instead, they both hear a depressing story from a voiceover recorded by an actor. When the voiceover ends, the game restarts, and they can try again. If the activist kills

the engineer, the ending is similar, only this time the engineer has full control of activist's body and there is a different story to hear.

The positive ending is harder to get and requires cooperation of all three players. The politician has a set of instructions which can enable a cooperative mode between engineer and activist. This cooperative mode changes the room in a way that activist can see the room by the guns ray, and the environment is suddenly harmful towards the engineer, who is relying on help from both politician and activist. In this mode the activist's gun can suddenly protect and guide the engineer towards a certain point, in which they all get to a good ending. After that, they appear in a distorted control room, which is no longer dangerous, and hear a positive story about the peaceful use of nuclear power.

2.2 Mapping the historical figures on the game characters

We integrated historical figures into our interactive project, adapting their behavior to both digital and physical interfaces. The engineering character's interaction took place via a 3D-printed PC interface, resembling machinery operation akin to engineers in power plants. This interface, designed with user-friendliness in mind, replicated an operator panel with oversized keys for non-gamers, as shown in Figure 4. Players guided a spider-like robot to extract power from reactors, which caused harm to the activist character. Using this setup can be seen in Figure 5.

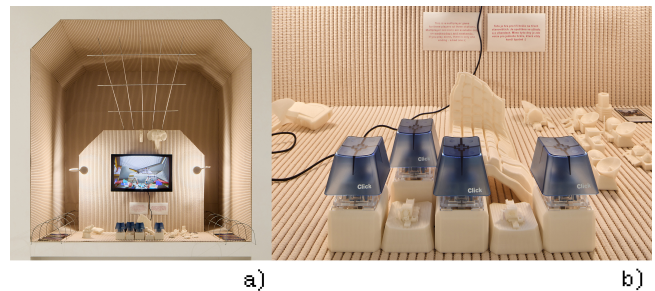


Figure 4: Engineering niche and custom-made interface a) Physical niche from distance b) Detail on four custom made keyboards. The inner switch is a product from NovelKeys LLC (© Vojtěch Radakulan)

The activist Character enters the virtual room by a VR headset embedded in a 3D printed cover. The niche is adjusted so that the visitor can sit and lean on a soft material. The headset is not mounted to the head and can be quickly put on as an old-fashioned theater mask. The activist can also use a gun, which is a controller embedded in a custom-made 3D printed case as shown in Figure 6. The idea behind this setup was to depict the activist as somebody who can cover his or her face and thus become masked while having a powerful weapon symbolized in a gun, which can also be misused.

The political character enters via a CCTV-like camera feed, using written instructions to encourage player cooperation. Operating solely through voice adds a challenge, portraying the political figure as an observer and communicator, rather than an active participant. The politician's setup includes a non-interactive computer screen,

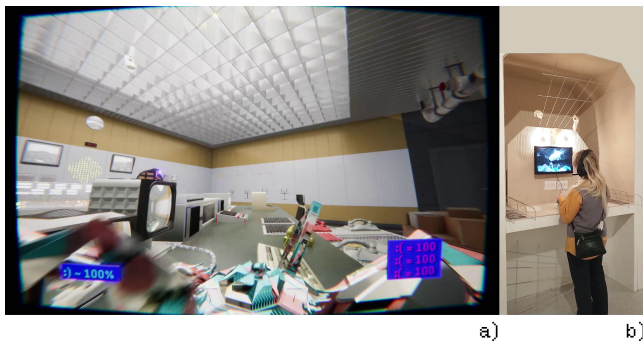


Figure 5: Engineering niche shows a) Screenshot from a PC when playing as an engineer b) A player interacting with the engineering niche (© Vojtěch Radakulan)

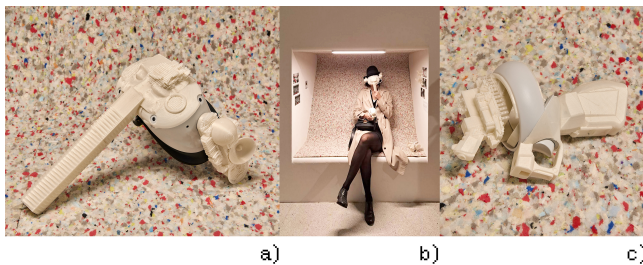


Figure 6: Installation of the activist niche with a) Oculus Quest 2 VR headset in a custom-made case as a mask b) The seating and visitor holding the headset c) Quest 2 controller embedded in a 3D printed case, which functions as the gun (© Vojtěch Radakulan)

emphasizing communication through microphone and listening to players as seen in Figure 7.



Figure 7: Installation of the politician niche, showing a) The niche from a distance b) Detail of the niche with the PC and instructions c) A player in process (© Vojtěch Radakulan)

3 USER EXPERIENCE AND EVALUATIONS

The installation was on display in a public gallery for four months. During that we did occasional audio and video recordings by common consent of visitors who came for the exhibition. We also recorded the number of visitors who were able to interact in any way with the installation and end up in any of three ends, as can

be seen in Figure 8. The diagram describing basic relationships of players can be seen in Figure 9.

The game was set up in a way that causing a bad ending was quite easy and a good ending needed effort. However, much better insight was gained by audio recordings, where some interesting interactions took place. The recording is attached to this document and its transcription is further analyzed in the next section.

count of trials	
any interaction	1253
204 bad ending total	
171 bad ending caused by the activist	
33 bad ending caused by the engineer	
28 good ending total	

Figure 8: During the four months display we recorded 1253 interactions, which resulted in 204 bad endings and 28 good endings. (© Vojtěch Radakulan)

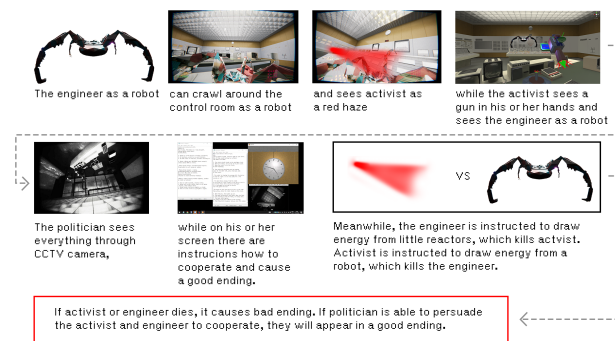


Figure 9: Schematic explanation of the basic relationships in the game. (© Vojtěch Radakulan)

3.1 How would it feel to be someone else? Keyword call outs describing behavior of players after switching roles

We present here four keyword call outs, which should illustrate our primary need – to foster empathy via changing roles of characters and understanding their positions. The recorded situations revolve around concerns about the wellbeing of other players, sharing valuable information of what to avoid or what to do, or sharing mistakes of the past trials.

(1) ***“You said you were feeling sick”***

In this example, one of the players mistakes a prerecorded message played during bad ending with a voice of the other player and is concerned about his or her well-being. During the bad ending the player who killed the other is being controlled by its victim, while they are not allowed to communicate by voice. Instead, they can only hear a prerecorded voiceover of a depressive story about nuclear power. When the bad ending ends, the players are moved back to the control room, where they can once again communicate by voice. In this case, one of the players thought that the voice of the actor was not recorded but was the voice of the other player. So, this is an excerpt of the communication after the team appears back in the room, as they are trying to figure out what they just went through:

Boy: *“What? You were next to me?”*

Girl: *“No, you were next to me! It was terrible. You said you were feeling sick. Didn’t I say something to you?”*

(By “feeling sick” girl refers to the actors prerecorded voice over, because she thinks that it was the voice of the boy)

Boy: *“What?”*

Girl: *“Didn’t the spider speak to you? You said to me that you were feeling sick!”*

Boy: *“No I didn’t say that!”*

Girl: *“You said that. The Glasses said that. The red haze said that! Now let us go and try something!”*

(2) ***“We already know what to do because I can remember the instructions!”***

In this situation, the team already played and switched roles many times, so one member was able to memorize all instructions and offered to guide less skilled players. Girl 1: *“Can we switch? I would like to be a politician character!”*

Girl 2: *“Ok but hurry up guys! No, no, beware. She cannot be a politician. She cannot read! And there are just written instructions!”*

Boy: But she does not have to read that. We already know what to do because I can remember the instructions!”

(3) ***“I want you to go through the drama I went through!”***

In this case, a player with the experience of bad ending wants the other player to go through it as well, so that it results in better understanding of both roles. A girl wants to kill a boy, so that he could go through the same drama she went through. This shows nicely a need for players to share the unpleasant experience of the bad ending, which could result in a mutual understanding of what to avoid. This is an excerpt of the conversation:

(Boy as an activist is shooting at the girl playing as an engineer, drawing the energy from her)

Girl: *“It looks like you are killing me a lot! Do not kill me! ... Ok, I get it, it is your need. So, what if we switch?”*

Boy: *“What?”*

Girl: *““Would you like to switch roles?”*

Boy: *“No”*

Girl: *“Please!”*

(Girl starts to draw energy from the boy)

Boy: *“Please do not kill me!”*

Girl: *“I will have to! I want you to go through the drama I went through!”*

Boy: *“No, no, no”*

(Team dies and goes to bad ending)

(4) ***“Just do not deplete it, as I did. When I was playing as your character, I was just randomly shooting around the room and depleted all the energy”***

This shows the case of giving advice resulting from past trials, thus showing emphatical behavior.

4 CONCLUSION

The main takeaway is the ability of the game to foster empathy in the players through the possibility of switching roles in the game. It can be shown on key call outs that the game enables players to show concerns about the well-being of other players, to share valuable information of what to do or avoid, or sharing mistakes of past trials. That is enabled by replaying and switching roles, thus resulting in a better understanding of the situation and task. Changing characters appears to be a suitable tool to tell a story where explaining opposing opinions is needed. User experience evaluations indicated that players showed empathetic behavior by showing concern for each other’s well-being, offering guidance, and looking to share their experiences to enhance mutual understanding. Through role-switching and replaying scenarios, players gained insight into the challenges and perspectives of others, enabling a more comprehensive understanding of historical events.

We also find out that when relying on an audio communication only, some people might misinterpret the voice of the actor as another player’s voice, which is something we didn’t expect, but it also results in concern for the wellbeing of others. We explored a way of attracting people to play with the digital interfaces by making them more approachable by custom made modifications to well-known interactions like playing with a keyboard or wearing a VR headset as an old-fashioned mask. This might open a new way of exhibiting interactive art in galleries, in that it promotes empathy and social interactions within a meaningful contextual background.

In conclusion, this study shows the ability of interactive digital media, specifically PC/VR gaming, to help empathetic engagement with historical events and differing viewpoints. By enabling players to step into the shoes of characters with conflicting opinions, the experience encourages nuanced understandings and dialogue around complex subjects, contributing to a broader goal of promoting empathy in modern society.

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